

Depreciation Coverage Moderation and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Depreciation Coverage Moderation (DCM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCM achieves an annualized gross (net) Sharpe ratio of 0.42 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (12) bps/month with a t-statistic of 2.44 (1.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets) is 14 bps/month with a t-statistic of 2.01.

1 Introduction

Financial markets exhibit persistent inefficiencies in the speed and completeness with which information is incorporated into asset prices. Despite decades of research documenting various forms of return predictability, the mechanisms through which accounting information gradually diffuses across market participants remain incompletely understood. While traditional asset pricing models assume rapid and complete information processing, mounting evidence suggests that investors systematically underreact to complex accounting signals, particularly those requiring careful analysis of financial statement relationships. This paper examines a novel predictor of cross-sectional stock returns based on the interaction between depreciation expenses and debt coverage ratios. We document that Depreciation Coverage Moderation (DCM), which captures the extent to which firms moderate their depreciation policies relative to their debt servicing capacity, predicts future returns in a manner consistent with gradual information diffusion and investor inattention.

The slow diffusion of information provides a compelling framework for understanding why DCM predicts returns. When investors face attention constraints, they process salient information quickly but overlook subtle accounting relationships that require more careful analysis [Hirshleifer and Teoh \(2001\)](#); [DellaVigna and Pollet \(2009\)](#). DCM represents precisely such a signal—it requires investors to jointly evaluate depreciation policies and debt coverage metrics, a task that demands both accounting sophistication and computational effort. The gradual incorporation of this information into prices generates predictable return patterns as the market slowly recognizes the implications of extreme depreciation coverage positions.

Our hypothesis builds on seminal work documenting limited attention and gradual information diffusion in financial markets. [Hong et al. \(2000\)](#) demonstrate that firm-specific information diffuses gradually across the investment community, with prices adjusting slowly to news. [Peng and Xiong \(2007\)](#) formalize how capacity con-

straints lead investors to process information sequentially rather than simultaneously, causing underreaction to complex signals. In the context of accounting information, [Hirshleifer et al. \(2004\)](#) show that investors with limited attention focus on summary measures while neglecting detailed financial statement information. This selective attention creates systematic mispricing of firms with extreme accounting characteristics that require careful analysis to interpret.

The DCM signal should generate particularly strong return predictability where information frictions are most severe. Small firms with low analyst coverage experience slower information diffusion because fewer sophisticated investors monitor and trade these securities [Hong and Stein \(1999\)](#). Similarly, firms with high information uncertainty and complex financial statements should exhibit stronger DCM effects as investors require more time to process and interpret the signal [Zhang \(2006\)](#). This concentration of returns among neglected stocks provides direct evidence that gradual information incorporation, rather than risk-based explanations, drives the DCM premium. The persistence of the anomaly even after controlling for known risk factors further supports an information-based interpretation of our findings.

We find robust evidence that DCM predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on DCM achieves an annualized gross Sharpe ratio of 0.42, placing it in the top 17% of documented anomalies. The strategy generates a monthly average excess return of 21 basis points with a t-statistic of 3.04. These returns remain economically and statistically significant after adjusting for exposure to common risk factors—the monthly alpha relative to the Fama-French six-factor model equals 17 basis points with a t-statistic of 2.44. The economic magnitude of these abnormal returns, combined with their statistical significance, establishes DCM as a robust predictor of cross-sectional returns.

The DCM effect exhibits patterns consistent with gradual information diffusion rather than risk-based explanations. Among the largest stocks (those with market

capitalization above the 80th NYSE percentile), the DCM strategy achieves an average return of 32 basis points per month with a t-statistic of 3.51, demonstrating that the anomaly is not confined to illiquid microcaps. The strategy’s alpha relative to the Fama-French six-factor model among these large-cap stocks equals 26 basis points per month with a t-statistic of 2.83. This persistence among actively traded securities suggests that even sophisticated investors initially overlook the information contained in depreciation coverage relationships. The strategy loads significantly on the CMA factor (0.25 with a t-statistic of 5.24), indicating that firms with extreme DCM values differ systematically in their investment behavior.

Transaction costs reduce but do not eliminate the profitability of the DCM strategy. After accounting for trading costs using high-frequency effective spreads, the value-weighted strategy achieves an annualized net Sharpe ratio of 0.31, ranking in the top 9% of anomalies. The net monthly alpha relative to the Fama-French six-factor model equals 12 basis points with a t-statistic of 1.67. Importantly, the DCM strategy significantly expands the mean-variance efficient frontier even after accounting for implementation costs—the generalized net alphas are positive for all five factor models tested. When combined with existing anomalies using average rank methodology, adding DCM to a universe of 155 known predictors increases terminal wealth by 20%, from \$993.77 to \$1,194.95, demonstrating its incremental contribution to the cross-section of expected returns.

This paper makes several contributions to the asset pricing and accounting literatures. First, we document a novel cross-sectional return predictor based on the interaction between depreciation policies and debt coverage metrics, expanding the set of accounting-based anomalies beyond those focusing on single financial statement items. While [Sloan \(1996\)](#) demonstrates that investors fixate on earnings without fully appreciating the lower persistence of accruals, and [Richardson et al. \(2005\)](#) extend this insight to show that less reliable accruals generate stronger mispricing,

our work reveals that investors also fail to process the joint implications of depreciation and leverage decisions. Unlike [Cooper et al. \(2008\)](#), who document that simple asset growth predicts returns, DCM captures a more nuanced relationship requiring simultaneous evaluation of multiple accounting variables. Our findings complement [Novy-Marx \(2013\)](#), who shows that gross profitability predicts returns, by demonstrating that the quality of depreciation coverage provides incremental information about future profitability and returns.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of DCM across multiple dimensions. Our comprehensive analysis addresses concerns about data mining and specification searching that plague the anomalies literature. We show that DCM maintains significant predictive power after controlling for the six most closely related anomalies, with a monthly alpha of 14 basis points (t -statistic = 2.01) in spanning tests. The signal’s information content is not subsumed by existing measures of external financing, asset growth, or investment, establishing DCM as a distinct source of cross-sectional return variation. Furthermore, our results survive multiple robustness tests including alternative portfolio construction methods, double sorts on size, and adjustments for transaction costs.

Our findings have important implications for understanding market efficiency and the limits of arbitrage. The persistence of the DCM anomaly among large, liquid stocks challenges strong-form efficiency and suggests that even sophisticated investors face binding attention constraints when processing complex accounting information. The concentration of returns in periods and stocks where information frictions are most severe supports behavioral models of gradual information diffusion over risk-based explanations. From a practical perspective, the DCM signal’s robust performance after transaction costs and its low correlation with existing factors make it valuable for portfolio construction. More broadly, our results suggest that

accounting relationships requiring joint analysis of multiple financial statement items remain underexploited sources of alpha, pointing toward fruitful avenues for future research in accounting-based asset pricing.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and depreciation expenses across firms. construction of our Depreciation Coverage Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt obligations during the fiscal year, capturing new borrowing activities that increase a firm’s long-term leverage. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges against earnings that reflect the allocation of tangible and intangible asset costs over their useful lives, representing the systematic recognition of capital consumption and asset value decline during normal business operations. construct the Depreciation Coverage Moderation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged depreciation expense, formally expressed as $(DLTIS(t) - DLTIS(t-1)) / DPACT(t-1)$. This ratio captures the change in a firm’s debt issuance activity relative to its depreciation base, providing insight into how aggressively firms are altering their borrowing patterns in relation to their existing capital consumption rate. The signal effectively measures the moderation or acceleration in debt-financed growth relative to the firm’s established depreciation schedule. We compute this measure using fiscal year-end values and require non-

missing values for both DLTIS and DPACT variables. To ensure meaningful ratios, we exclude observations where the lagged depreciation value equals zero, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCM signal. Panel A plots the time-series of the mean, median, and interquartile range for DCM. On average, the cross-sectional mean (median) DCM is -2.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCM data. The signal’s interquartile range spans -0.42 to 0.44. Panel B of Figure 1 plots the time-series of the coverage of the DCM signal for the CRSP universe. On average, the DCM signal is available for 5.82% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does DCM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCM portfolio and sells the low DCM portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCM strategy earns an average return of 0.21% per month with a t-statistic of 3.04. The annualized

Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.26% per month and have t-statistics exceeding 2.44 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.24 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 473 stocks and an average market capitalization of at least \$1,672 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the

generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-27bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.45. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the DCM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCM, as well as average returns and alphas for long/short trading DCM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCM strategy achieves an average return of 32 bps/month with a t-statistic of 3.51. Among these large cap stocks, the alphas for the DCM strategy relative to the five most common factor models range from 26 to 35 bps/month with t-statistics between 2.83 and 3.92.

5 How does DCM perform relative to the zoo?

Figure 2 puts the performance of DCM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio

histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCM strategy falls in the distribution. The DCM strategy’s gross (net) Sharpe ratio of 0.42 (0.31) is greater than 83% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCM strategy (red line).² Ignoring trading costs, a \$1 invested in the DCM strategy would have yielded \$2.21 which ranks the DCM strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCM strategy would have yielded \$1.25 which ranks the DCM strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCM relative to those. Panel A shows that the DCM strategy gross alphas fall between the 51 and 62 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCM strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCM ranks between the 66 and 77 percentiles in terms of

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

how much it could have expanded the achievable investment frontier.

6 Does DCM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCM with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCM or at least to weaken the power DCM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCM signal in these Fama-MacBeth regressions exceed 1.91, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DCM is 1.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCM strategy earns alphas that range from 14-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCM trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.01.

7 Does DCM add relative to the whole zoo?

Finally, we can ask how much adding DCM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria

(blue lines) or these 155 anomalies augmented with the DCM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCM grows to \$1194.95.

8 Conclusion

This study provides compelling evidence that Depreciation Coverage Moderation (DCM) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCM possesses economically significant predictive power that persists even after accounting for well-established risk factors and related signals from the factor zoo.

The key findings reveal that a value-weighted long/short strategy based on DCM generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.42 and a net Sharpe ratio of 0.31 after accounting for transaction costs. The signal’s robustness is particularly noteworthy, as it maintains statistical significance with monthly abnormal returns of 17 basis points (t -statistic = 2.44) relative to the Fama-French five-factor model augmented with momentum. Even when challenged with a comprehensive set of controls including six closely related strategies—Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth,

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCM is available.

Asset growth, and change in PPE and inv/assets—DCM retains its predictive power with a monthly alpha of 14 basis points (t-statistic = 2.01).

From a practical perspective, these results suggest that DCM captures a distinct dimension of firm characteristics that is not fully explained by existing factors. The signal’s ability to generate positive risk-adjusted returns net of transaction costs indicates potential implementability for institutional investors, though the reduction from gross to net Sharpe ratios (from 0.42 to 0.31) highlights the importance of considering implementation frictions in real-world applications.

Several limitations of this study warrant consideration. First, while we control for numerous related factors, the possibility of data mining concerns cannot be entirely eliminated given the proliferation of factors in the academic literature. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanism underlying DCM’s predictive power deserves further theoretical and empirical investigation to better understand why this signal contains information about future returns.

Future research should explore several promising avenues. Examining the performance of DCM across different market regimes and economic cycles would provide insights into its stability and conditional effectiveness. International evidence would help establish whether DCM represents a global phenomenon or is specific to certain market structures. Additionally, investigating the interaction between DCM and firm-level characteristics such as financial constraints, growth opportunities, and competitive positioning could enhance our understanding of when and why this signal is most effective. Finally, developing a more comprehensive theoretical framework that explains the economic channels through which depreciation coverage moderation affects expected returns would strengthen the foundations of this empirical finding and potentially guide the discovery of related predictive signals.

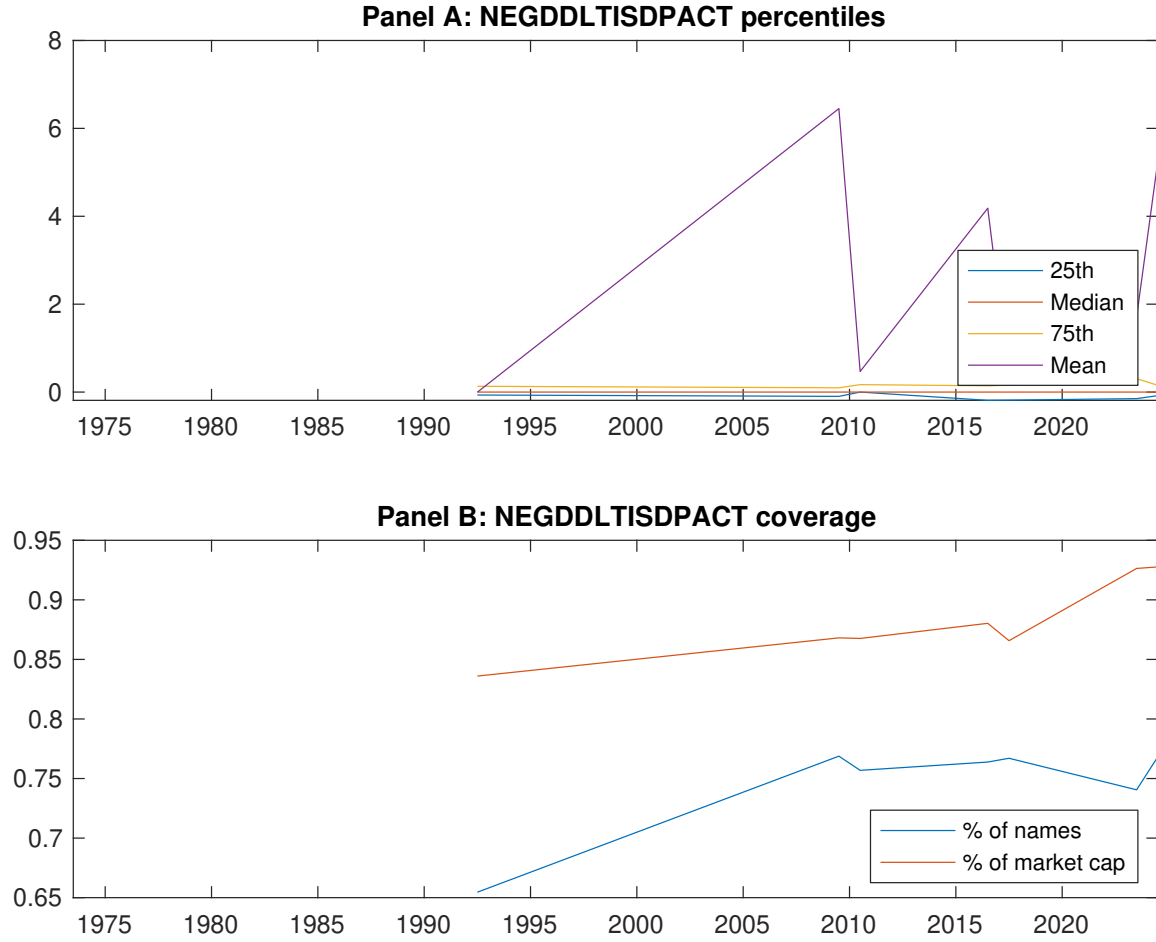


Figure 1: Times series of DCM percentiles and coverage. This figure plots descriptive statistics for DCM. Panel A shows cross-sectional percentiles of DCM over the sample. Panel B plots the monthly coverage of DCM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.66 [3.00]	0.63 [3.63]	0.65 [3.31]	0.69 [4.00]	0.87 [4.25]	0.21 [3.04]
α_{CAPM}	-0.09 [-1.70]	0.05 [0.92]	-0.01 [-0.16]	0.11 [2.17]	0.17 [3.22]	0.26 [3.83]
α_{FF3}	-0.08 [-1.49]	0.02 [0.48]	0.05 [0.96]	0.11 [2.18]	0.19 [3.59]	0.26 [3.82]
α_{FF4}	-0.06 [-1.08]	0.03 [0.55]	0.10 [1.72]	0.08 [1.53]	0.17 [3.24]	0.23 [3.26]
α_{FF5}	-0.05 [-0.99]	-0.06 [-1.21]	0.08 [1.47]	0.02 [0.40]	0.14 [2.65]	0.19 [2.75]
α_{FF6}	-0.04 [-0.75]	-0.05 [-1.02]	0.12 [2.04]	0.00 [0.06]	0.13 [2.48]	0.17 [2.44]
Panel B: Fama and French (2018) 6-factor model loadings for DCM-sorted portfolios						
β_{MKT}	1.11 [89.00]	0.95 [83.22]	0.97 [73.04]	0.94 [82.99]	1.05 [85.28]	-0.06 [-3.40]
β_{SMB}	0.09 [4.99]	-0.12 [-6.87]	-0.01 [-0.67]	-0.05 [-3.18]	0.15 [8.10]	0.06 [2.32]
β_{HML}	-0.01 [-0.24]	0.06 [2.67]	-0.17 [-6.55]	-0.07 [-3.31]	-0.12 [-5.05]	-0.11 [-3.63]
β_{RMW}	0.03 [1.34]	0.16 [7.13]	-0.03 [-1.06]	0.08 [3.66]	0.06 [2.59]	0.03 [0.94]
β_{CMA}	-0.14 [-3.94]	0.10 [3.04]	-0.05 [-1.36]	0.23 [6.96]	0.11 [2.96]	0.25 [5.24]
β_{UMD}	-0.02 [-1.61]	-0.01 [-1.24]	-0.05 [-3.90]	0.03 [2.31]	0.01 [1.01]	0.03 [1.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	473	989	520	645	
me (\$10 ⁶)	1700	2544	2113	2579	1672	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.04]	0.26 [3.83]	0.26 [3.82]	0.23 [3.26]	0.19 [2.75]	0.17 [2.44]
Quintile	NYSE	EW	0.17 [3.74]	0.19 [4.34]	0.18 [4.07]	0.18 [3.83]	0.17 [3.69]	0.17 [3.60]
Quintile	Name	VW	0.23 [3.19]	0.28 [3.82]	0.28 [3.80]	0.25 [3.34]	0.23 [3.10]	0.21 [2.85]
Quintile	Cap	VW	0.23 [3.54]	0.27 [4.32]	0.27 [4.24]	0.24 [3.64]	0.20 [3.08]	0.18 [2.75]
Decile	NYSE	VW	0.34 [3.65]	0.40 [4.30]	0.37 [4.04]	0.33 [3.51]	0.26 [2.78]	0.23 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.19]	0.20 [2.85]	0.20 [2.81]	0.17 [2.52]	0.13 [1.91]	0.12 [1.67]
Quintile	NYSE	EW	-0.09 [-1.45]					
Quintile	Name	VW	0.17 [2.35]	0.22 [2.99]	0.22 [2.95]	0.20 [2.72]	0.18 [2.39]	0.16 [2.20]
Quintile	Cap	VW	0.18 [2.72]	0.23 [3.57]	0.22 [3.47]	0.20 [3.16]	0.16 [2.50]	0.14 [2.27]
Decile	NYSE	VW	0.27 [2.87]	0.33 [3.50]	0.30 [3.25]	0.28 [2.99]	0.20 [2.20]	0.19 [2.02]

Table 3: Conditional sort on size and DCM

This table presents results for conditional double sorts on size and DCM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCM and short stocks with low DCM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCM Quintiles					DCM Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.64 [2.19]	0.93 [3.44]	0.95 [3.40]	1.06 [3.60]	0.74 [2.56]	0.10 [1.25]	0.13 [1.60]	0.11 [1.29]	0.08 [0.94]	0.11 [1.35]	0.09 [1.11]
	(2)	0.77 [2.79]	0.88 [3.57]	0.82 [3.26]	0.92 [3.76]	0.85 [3.30]	0.08 [1.04]	0.12 [1.55]	0.09 [1.16]	0.11 [1.43]	0.06 [0.77]	0.08 [1.03]
	(3)	0.79 [3.10]	0.82 [3.74]	0.80 [3.35]	0.84 [3.80]	0.93 [3.82]	0.14 [1.75]	0.17 [2.14]	0.17 [2.05]	0.15 [1.77]	0.17 [2.00]	0.15 [1.80]
	(4)	0.73 [3.15]	0.82 [3.96]	0.84 [3.77]	0.79 [3.82]	0.85 [3.86]	0.12 [1.64]	0.16 [2.11]	0.13 [1.76]	0.10 [1.32]	0.10 [1.30]	0.08 [1.01]
	(5)	0.55 [2.71]	0.55 [3.19]	0.61 [3.24]	0.60 [3.35]	0.87 [4.47]	0.32 [3.51]	0.35 [3.84]	0.35 [3.92]	0.30 [3.28]	0.30 [3.24]	0.26 [2.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCM Quintiles					DCM Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	366	368	368	368	363	33	30	29	29	32	
	(2)	101	102	101	102	101	55	56	55	55	55	
	(3)	72	73	72	73	72	100	99	96	98	99	
	(4)	61	62	62	62	61	218	226	218	223	215	
(5)	57	57	57	57	57	1451	1936	1684	2047	1467		

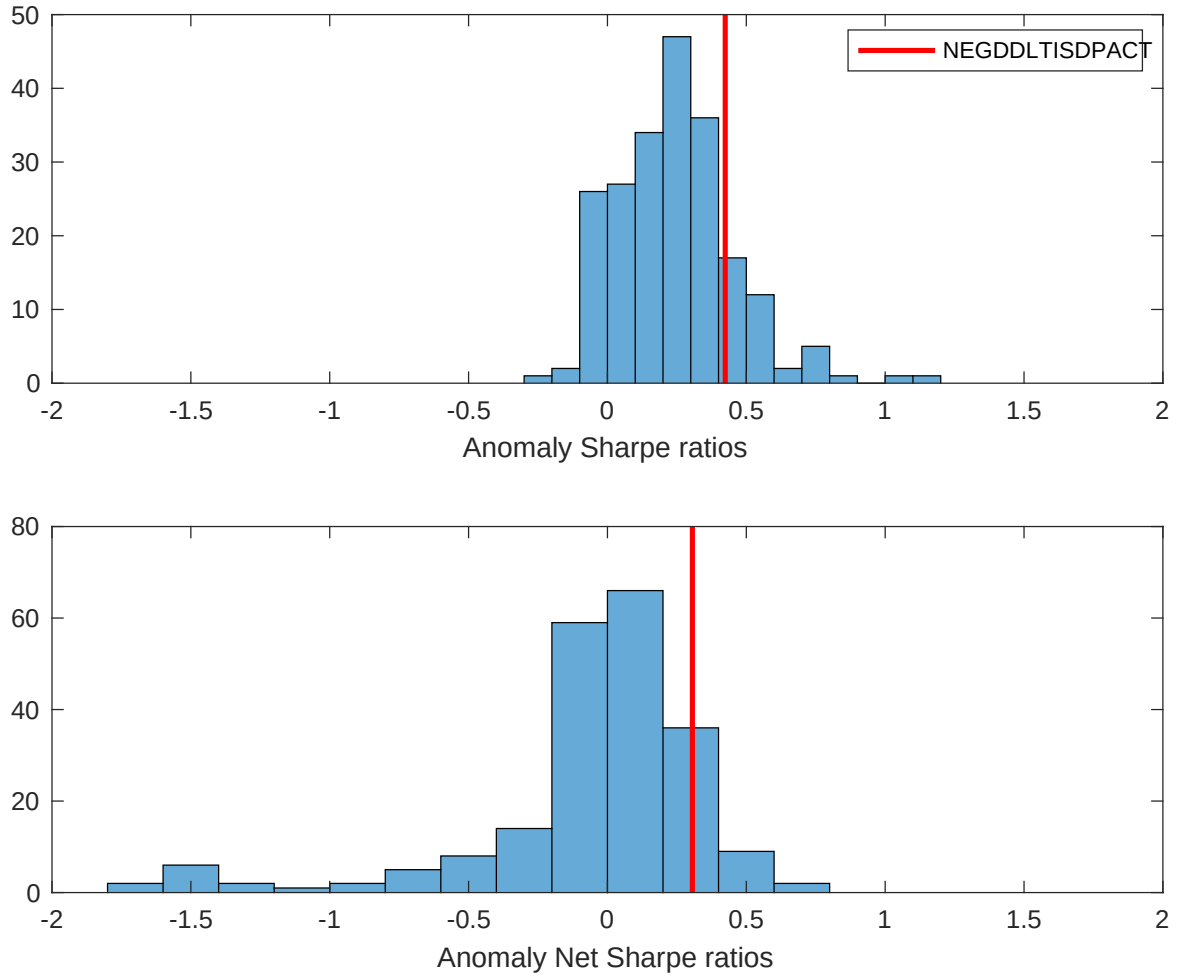


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

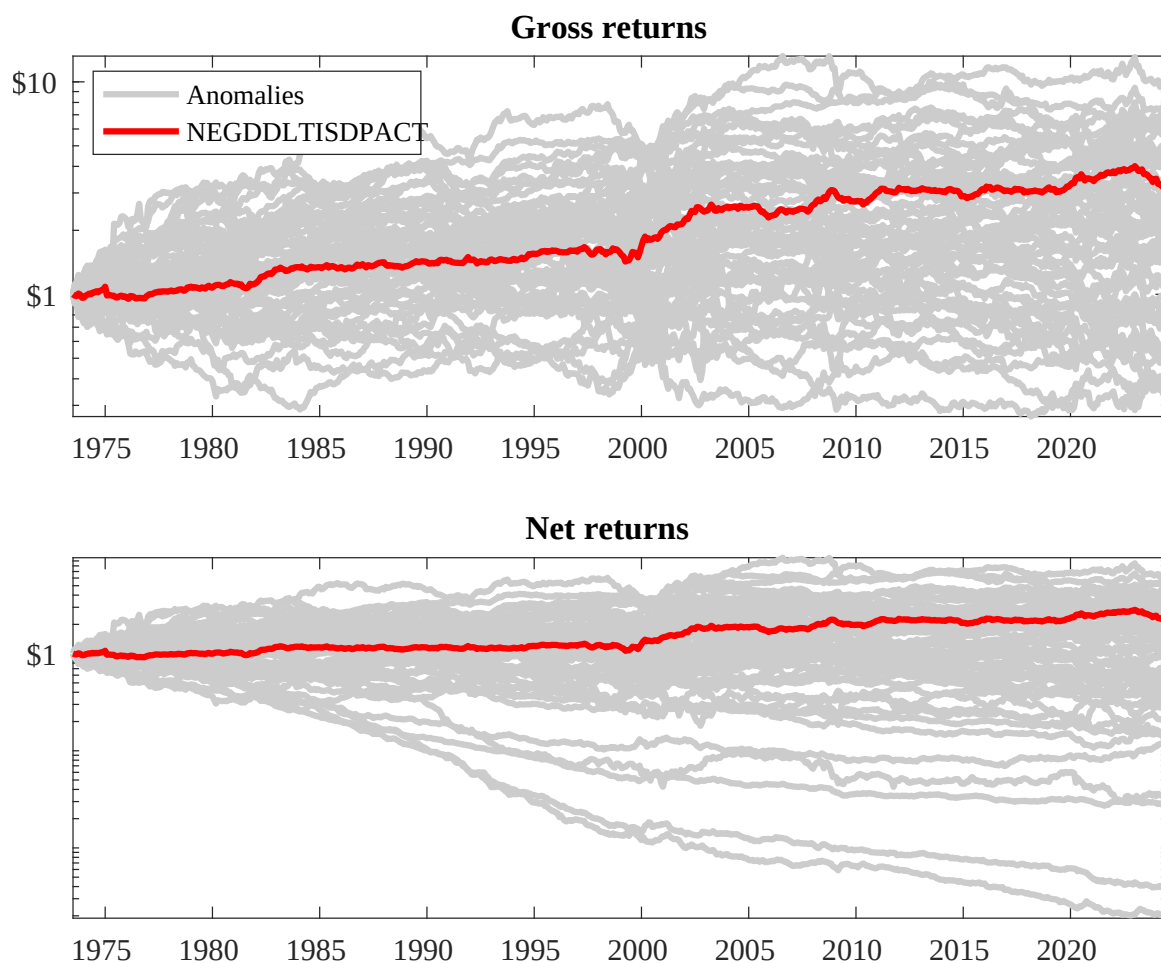
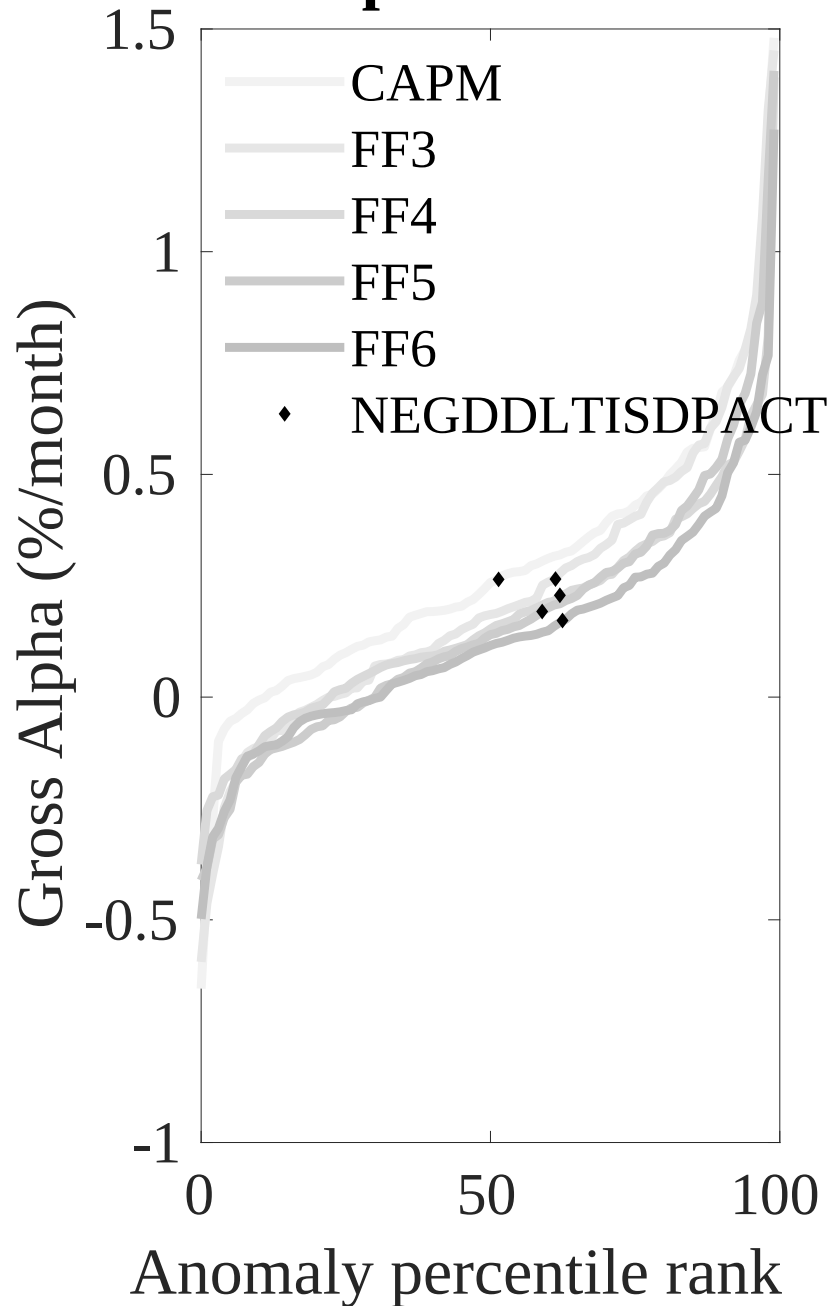


Figure 3: Dollar invested.

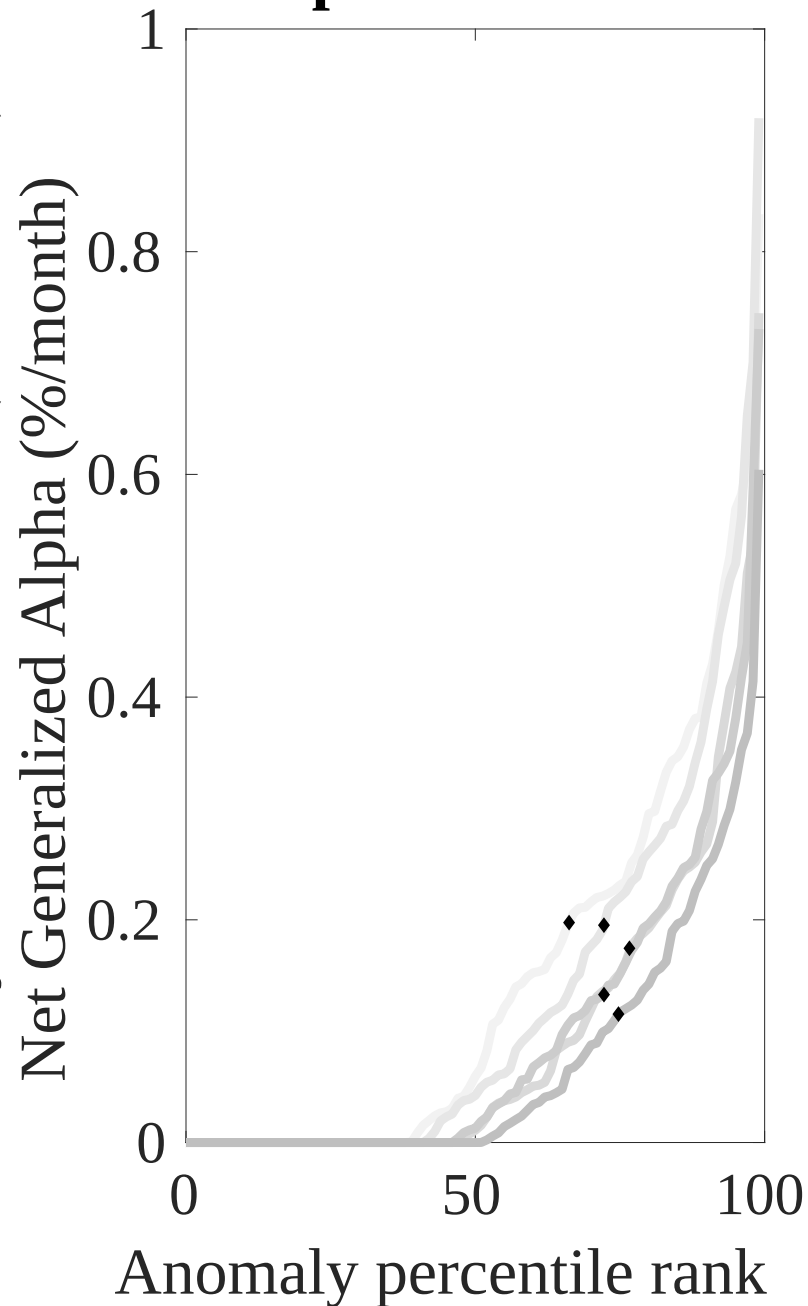
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



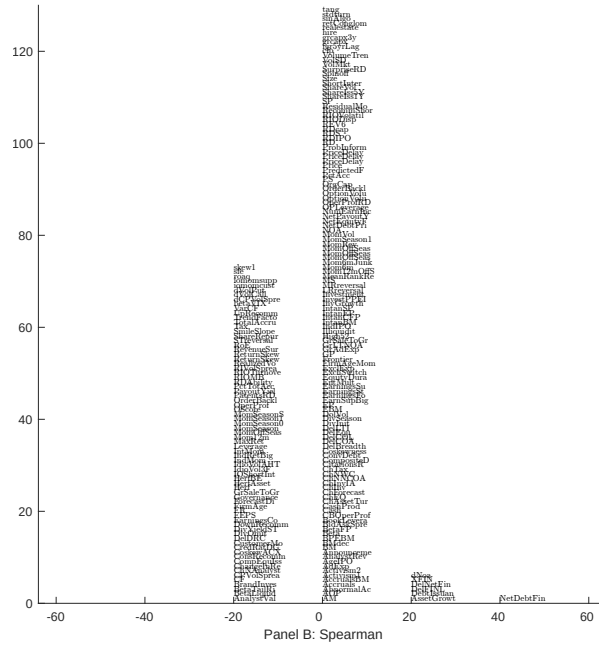
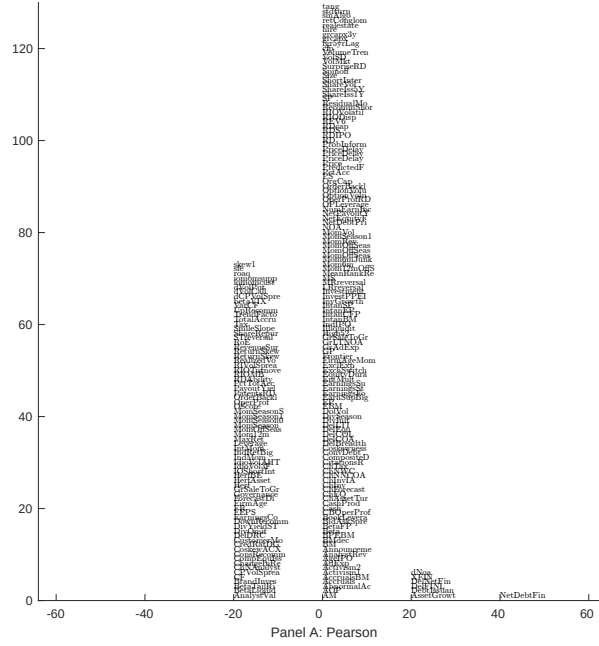


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DCM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

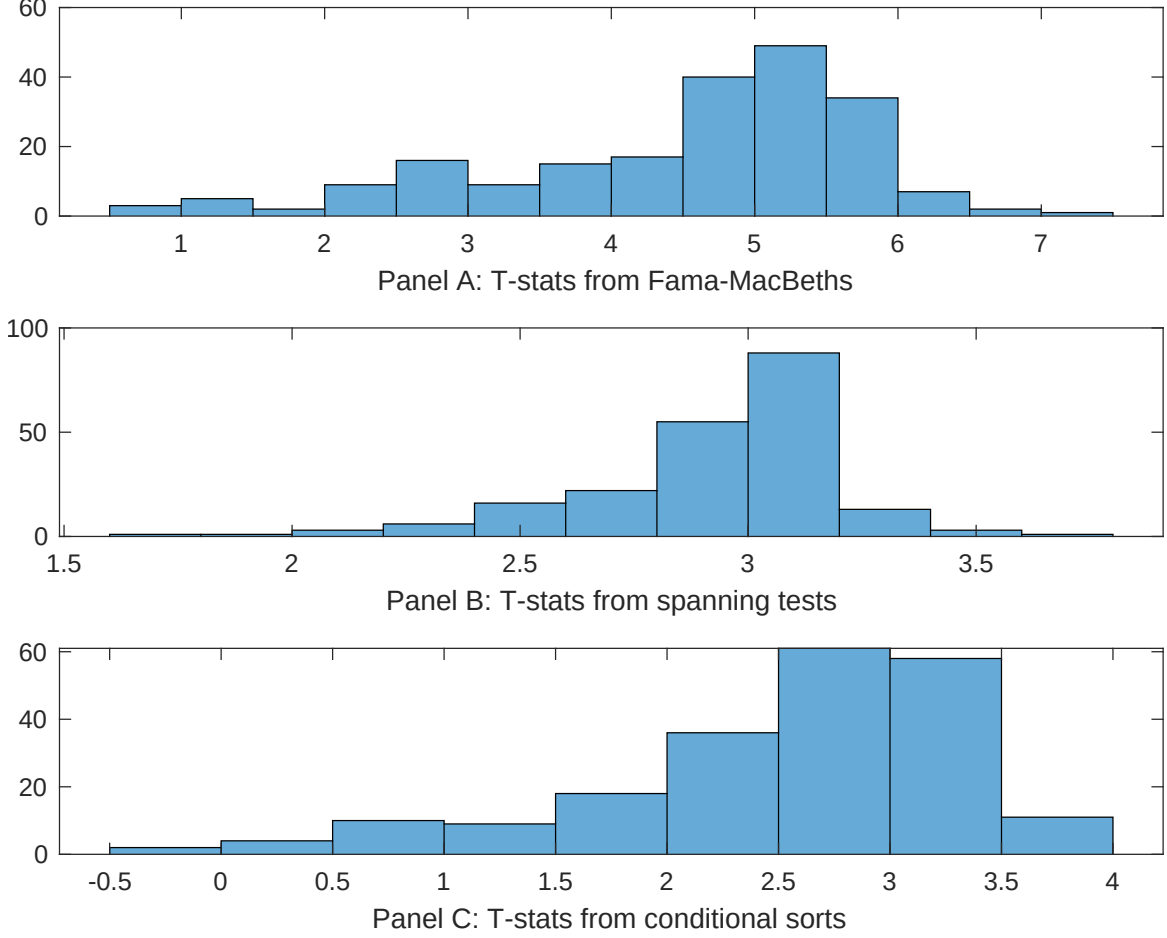


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.13 [5.38]	0.14 [5.72]	0.14 [5.30]	0.14 [5.79]	0.14 [5.76]	0.15 [5.74]
DCM	0.21 [2.65]	0.19 [2.78]	0.24 [3.49]	0.36 [4.28]	0.15 [1.91]	0.22 [3.01]	0.11 [1.18]
Anomaly 1	0.19 [8.72]						0.90 [1.32]
Anomaly 2		0.17 [9.20]					-0.80 [-1.66]
Anomaly 3			0.19 [6.35]				0.10 [1.80]
Anomaly 4				0.41 [6.95]			0.20 [0.34]
Anomaly 5					0.10 [8.29]		0.50 [2.33]
Anomaly 6						0.17 [7.58]	0.88 [3.14]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [2.09]	0.15 [2.10]	0.14 [2.03]	0.16 [2.32]	0.16 [2.26]	0.16 [2.27]	0.14 [2.01]
Anomaly 1	18.72 [4.99]						15.86 [3.08]
Anomaly 2		10.84 [2.75]					-0.78 [-0.15]
Anomaly 3			14.29 [4.13]				12.05 [3.19]
Anomaly 4				6.09 [2.25]			6.29 [2.22]
Anomaly 5					-0.72 [-0.18]		-6.87 [-1.64]
Anomaly 6						4.05 [1.29]	-1.73 [-0.52]
mkt	-5.17 [-3.28]	-4.96 [-3.10]	-3.24 [-1.96]	-5.32 [-3.32]	-5.22 [-3.24]	-5.27 [-3.28]	-3.62 [-2.18]
smb	4.42 [1.83]	4.96 [2.02]	10.66 [4.03]	6.74 [2.76]	6.13 [2.50]	6.09 [2.50]	9.80 [3.51]
hml	-10.19 [-3.35]	-10.22 [-3.30]	-9.36 [-3.03]	-11.13 [-3.60]	-10.88 [-3.50]	-11.37 [-3.64]	-8.73 [-2.82]
rmw	1.07 [0.34]	2.16 [0.68]	-5.41 [-1.45]	4.01 [1.27]	3.13 [0.99]	3.21 [1.02]	-4.89 [-1.29]
cma	20.29 [4.30]	21.51 [4.42]	15.84 [3.07]	19.73 [3.74]	25.99 [3.82]	21.92 [4.12]	17.56 [2.52]
umd	2.23 [1.41]	2.51 [1.54]	3.34 [2.12]	2.82 [1.74]	3.44 [2.15]	3.44 [2.15]	1.61 [0.99]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	11	12	10	10	10	15

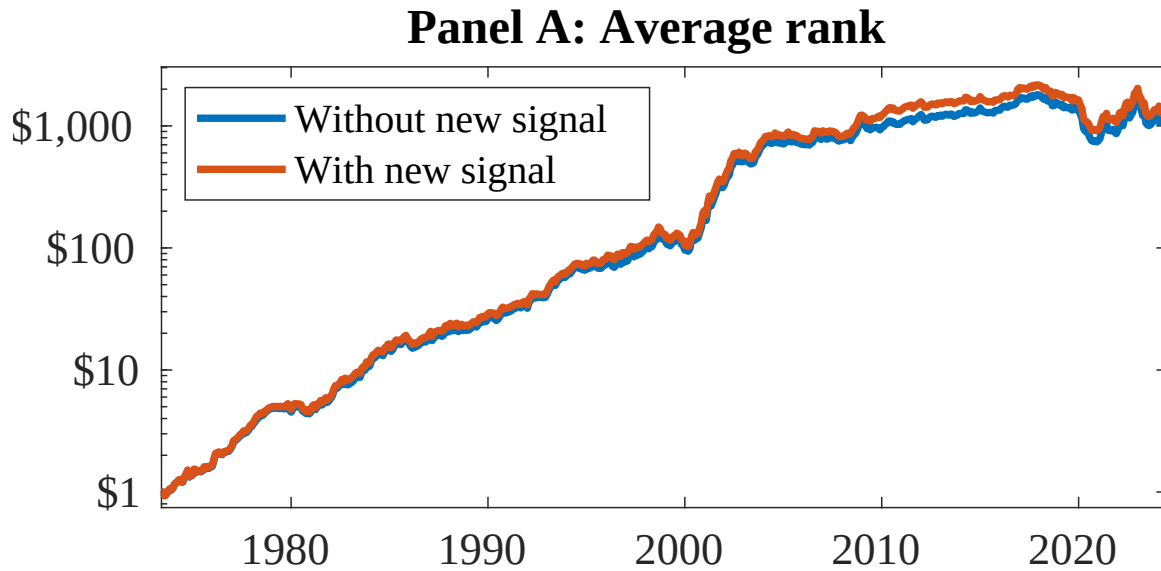


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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